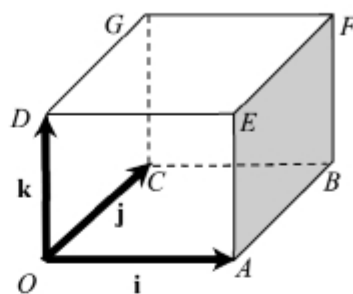


 Tutorial 7A Solutions

1. In the diagram, $OABCDEFG$ is a cube of side 1 unit.

Find the position vectors of

- (a) the point A ,
 (b) the point B ,
 (c) the point F ,
 (d) the midpoint of BD ,
 (e) the point that divides AG in the ratio 4 : 3.



Solution

(a) $\vec{OA} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ (b) $\vec{OB} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ (c) $\vec{OF} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

(d) By ratio theorem, $\vec{OM} = \frac{\vec{OB} + \vec{OD}}{2} = \frac{1}{2} \left[\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Note that $\vec{OM} \neq \frac{1}{2} \vec{BD}$.

In fact, $\vec{BM} = \frac{1}{2} \vec{BD}$.

(e) By ratio theorem, $\vec{OP} = \frac{4\vec{OG} + 3\vec{OA}}{7} = \frac{1}{7} \left[4 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right] = \frac{1}{7} \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix}$

2. The points A, B, C, D have coordinates $(0, 1, 3)$, $(4, 5, -5)$, $(-3, 0, -1)$ and $(7, 5, 4)$ respectively.

- (a) Find the length of the line segment AB .
 (b) Justify whether $ABCD$ is a parallelogram.
 (c) The point P divides AB in the ratio 1 : 3. Find $\vec{OP}$ and hence show that P also divides CD in the ratio 2 : 3.

Solution

(a) $\vec{AB} = \vec{OB} - \vec{OA} = \begin{pmatrix} 4 \\ 5 \\ -5 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ -8 \end{pmatrix}$

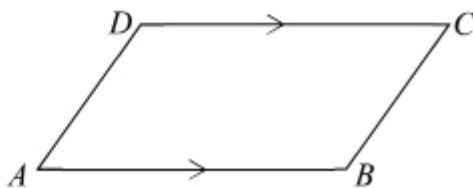
$$|\vec{AB}| = \left| \begin{pmatrix} 4 \\ 4 \\ -8 \end{pmatrix} \right| = 4 \left| \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right| = 4\sqrt{1^2 + 1^2 + 2^2} = 4\sqrt{6} \text{ units}$$

(b) $\vec{AB} = \begin{pmatrix} 4 \\ 4 \\ -8 \end{pmatrix}$

If $ABCD$ is a parallelogram, then there are two pairs of equal vectors (same length, same direction) $\vec{AB} = \vec{DC}$ and $\vec{AD} = \vec{BC}$.

$$\vec{DC} = \vec{OC} - \vec{OD} = \begin{pmatrix} -3 \\ 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 7 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} -10 \\ -5 \\ -5 \end{pmatrix}$$

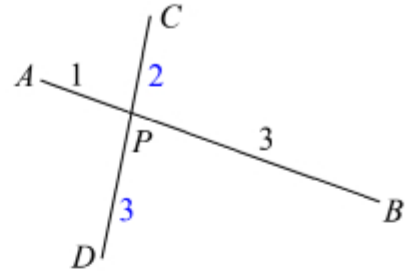
Since $\vec{AB} \neq \vec{DC}$, $ABCD$ is not a parallelogram.



(c) Given $\vec{OP} = \frac{3\vec{OA} + \vec{OB}}{4} = \frac{1}{4} \left[3 \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \\ -5 \end{pmatrix} \right] = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$

$\vec{CP} = \vec{OP} - \vec{OC} = 2 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ and $\vec{PD} = \vec{OD} - \vec{OP} = 3 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

Thus, $\vec{CP} = \frac{2}{3} \vec{PD} \Rightarrow CP : PD = 2 : 3$ (shown)



Note:

1. Cannot just find $|\vec{CP}| = 2\sqrt{6}$, $|\vec{PD}| = 3\sqrt{6}$ and conclude that $\frac{|\vec{CP}|}{|\vec{PD}|} = \frac{2\sqrt{6}}{3\sqrt{6}}$ and hence

$CP : PD = 2 : 3$. This does not show C, P and D are collinear. Need to show that $\vec{CP} = \frac{2}{3} \vec{PD}$.

2. It is incorrect to write $\frac{\vec{CP}}{\vec{PD}} = \frac{2}{3}$, since there is no division of vectors.

3. Can also show $\vec{CP} = \frac{2}{5} \vec{CD}$.

3. [HCI06/H2 Promo/Q8(modified)]

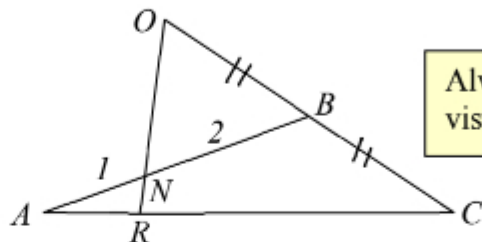


The position vectors of the points, A and B , relative to the origin O are $\mathbf{a} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ respectively. The point C lies on OB produced such that $OC = 2OB$.

- Find the position vector of the point N on AB such that $AN : NB = 1 : 2$.
- Given that the point R lies on ON produced such that the points A , R and C are collinear, find the position vector of the point R relative to the origin O .
- Given that the point P lies on BA produced such that the ratio of $BA : AP$ is $2 : 5$, find the coordinates of P .

Solution

$$(i) \quad \vec{ON} = \frac{\vec{OB} + 2\vec{OA}}{3} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$



Always draw diagram to visualise vectors involved.

- Method 1: (expressing $\vec{OR}$ in two different ways)

Since R lies on ON produced, $\vec{OR} = k\vec{ON} = k \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \dots (1)$

Also, since A , R and C are collinear,

$$\vec{AR} = t\vec{AC} = t(\vec{OC} - \vec{OA}) = t(2\vec{OB} - \vec{OA})$$

$$\vec{AR} = t \left[\begin{pmatrix} 4 \\ -6 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right] = t \begin{pmatrix} 2 \\ -6 \\ 1 \end{pmatrix}$$

$$\vec{OR} = \vec{OA} + \vec{AR} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 2 \\ -6 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+2t \\ -6t \\ 1+t \end{pmatrix} \dots (2)$$

$$(1) = (2), \quad k \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+2t \\ -6t \\ 1+t \end{pmatrix}$$

$$\begin{cases} 2k = 2+2t & \dots (3) \\ -k = -6t & \dots (4) \\ k = 1+t & \dots (5) \end{cases}$$

Note that $(3) = (5)$.

Hence solving (4) and (5), $k = \frac{6}{5}$, $t = \frac{1}{5}$

$$\therefore \vec{OR} = \frac{6}{5} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

Express $\vec{OR}$ in two different forms, equate the two forms and solve for k or t to find $\vec{OR}$.

If we have more equations than unknowns and the equations are all different, it is important to substitute answers into remaining unused equations to check for consistency of the answers. If the answers do not satisfy any of the remaining equations, then the answers are inconsistent and we conclude that there is no solution. In this case, since $(3) = (5)$, there are only 2 equations and 2 unknowns.

